

Generation Interconnection System Impact Study Report

For Queue Project Q17-0816

Igo 138 kV
March 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0816. The Interconnection Customer proposes a 46.7 MW solar photovoltaic facility interconnecting at the Igo 138 kV substation in the Virginia area. The request was received on June 27, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 279 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,774,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Igo-Thalia 138kV Line 1: 108.6% loading (333 MVA) following the outage of the Thalia-Igo 138kV Cir. 2; DFAX = 50.3%.

Igo Substation-Thalia 138kV Line: 103.6% loading (349 MVA) following the outage of the Thalia-Igo 138kV; DFAX = 50.3%.

Hood Mesa-Thalia Substation 138kV L: 101.5% loading (254 MVA) following the outage of the Thalia-Lynnhaven 138kV; DFAX = 46.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Igo-Thalia 138kV Line 1	Reconductor line	\$2,232,000	72% (\$1,600,000)
Igo Substation-Thalia 138kV Line	Reconductor line	\$1,580,000	72% (\$1,133,000)
Thalia-Hood Mesa 138kV	Reconductor line	\$1,405,000	74% (\$1,041,000)
	Total allocated to project		\$3,774,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.